Merkato Mixed-Use Wholesale-Trade Complex
Site-Suitability Feasibility Study

Planning-stage screening · Addis Ababa, Ethiopia · 2026-07-14 · CRS EPSG:32637 · 20 m analysis grid

1. Executive summary

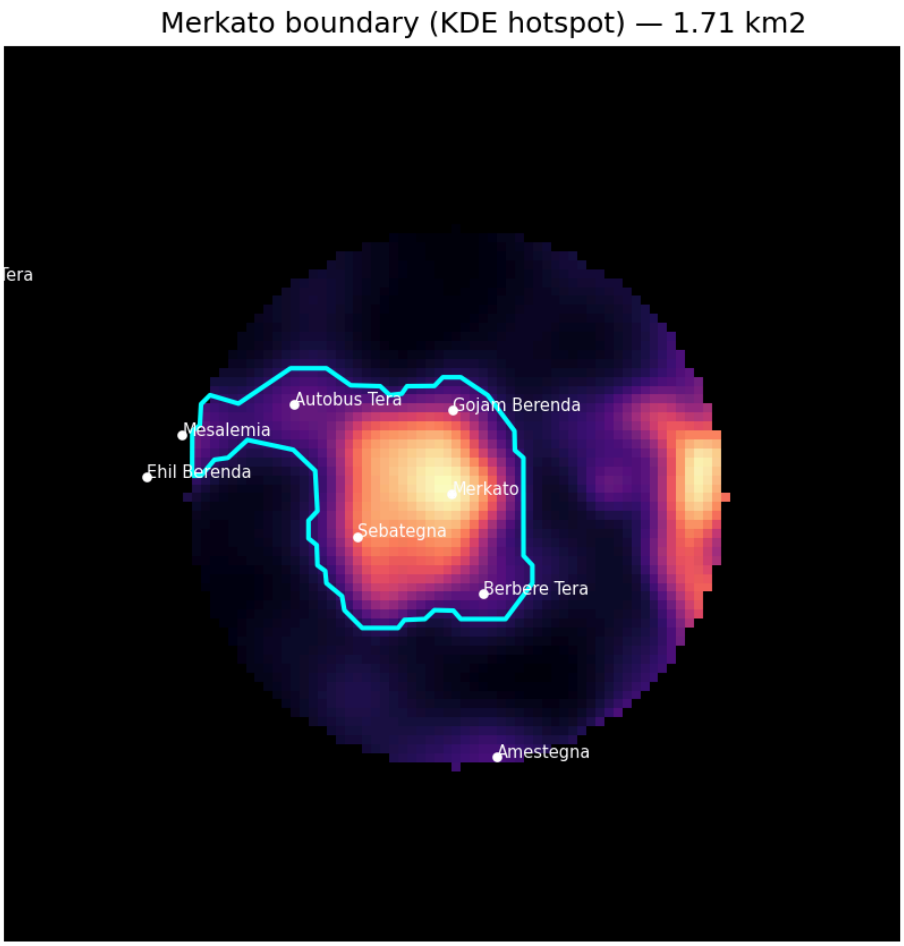
This study delineates the Merkato commercial area (1.71 km²) and screens it with a nine-criterion weighted-overlay (MCDA) model for parcels ≥ 5,000 m² able to host a mixed-use wholesale complex with three basement levels, automated logistics and a data-centre node. Weighting is **strongly market-forward per client direction** (trade gravity 0.32, dominant weight) and the market criterion is built around the observed commercial-POI hotspot. **Per client direction the headline deliverable is not a parcel shortlist but a siting heat map**: the weighted suitability surface, continuously graded down near satellite-measured tall buildings (penalty from 8 m, zero at ≥ 15 m) and large footprints (≥ 800 m²), with hard exclusions removed. The hottest areas are vectorised into **30 priority-A zones (17.8 ha, nearest 58 m from the core anchor)** and 25 priority-B zones (13.7 ha). These are investigation areas for groundwork — cadastre verification, occupancy census, geotechnical survey, government consent — after which specific parcels can be pinpointed. Within 400 m of the anchor, 628 of 1,225 grid cells carry measured heights ≥ 12 m and 341 large-footprint structures, so near-core opportunities are pockets, not blocks. Three illustrative parcels (§6.1) show what the zones can yield; they are calibration examples, not proposals. The base and displacement-weighted alternative surfaces correlate at r ≈ 0.98.

2. Objective and constraints

- Parcel ≥ 5,000 m², buildable geometry (compactness ≥ 0.25, ≤ 6 ha single parcel).
- Three basement levels → sensitivity to slope, drainage and flood risk (geotechnical survey mandatory before commitment).
- Automated logistics + data-centre node → electrical capacity proximity.
- Wholesale core → arterial/freight access; high footfall → transit access.
- Explicit displacement minimisation → vacant / low-rise / dilapidated fabric preferred and weighted highest.

3. Study-area delineation

The boundary was derived, not assumed: a 50 m kernel-density surface was built from (a) OSM commercial POIs (870 points used), (b) market/commerce polygons (CB-* codes) of the 2022 Structural Plan land-use layer, and (c) named sub-market anchor points (Merkato, Sebategna, Berbere Tera, Gojam Berenda, Autobus Tera, Mesalemia…). The 18% iso-intensity contour containing the Merkato point was retained: **1.71 km²**.



4. Data sources and vintage

Layer	Source	Vintage	Role
Land use (11,363 polys citywide)	AA Structural Plan First-Term Amendment (client-provided FileGDB)	June 2022	Zoning compatibility, exclusions, boundary
Road network & transport	AA Structural Plan (client-provided FileGDB)	June 2022	Freight/transit access, terminals, LRT
Building height regulation	AA BH Regulation Amendment (client-provided FileGDB)	June 2022	Height-headroom criterion
Roads, buildings, POIs, power	OpenStreetMap via Overpass	Live, July 2026	Hierarchy, POI density, heights (570 tagged)
Building footprints (131,701)	VIDA combined Google Open Buildings v3 / Microsoft / OSM	2023 release	Coverage / vacancy criterion
Building heights (satellite-measured, 0.5 m grid)	Google Open Buildings 2.5D Temporal, building_height + building_presence bands	June 2023	Low-rise verification; high-rise exclusion in vacancy criterion and site filters
Site imagery audit	Esri World Imagery (Maxar), zoom 18	current basemap	Visual verification of each shortlisted site
DEM 30 m	Copernicus GLO-30 (AWS COG)	2021 (GLO-30 2021 release)	Slope, drainage proxy
Population 100 m	WorldPop 2020 UN-adjusted constrained	2020	Catchment density

All layers were validated, repaired (buffer-0), reprojected to EPSG:32637 and clipped to the study bbox. Provided layers arrived in EPSG:20137 (Adindan / UTM 37N).

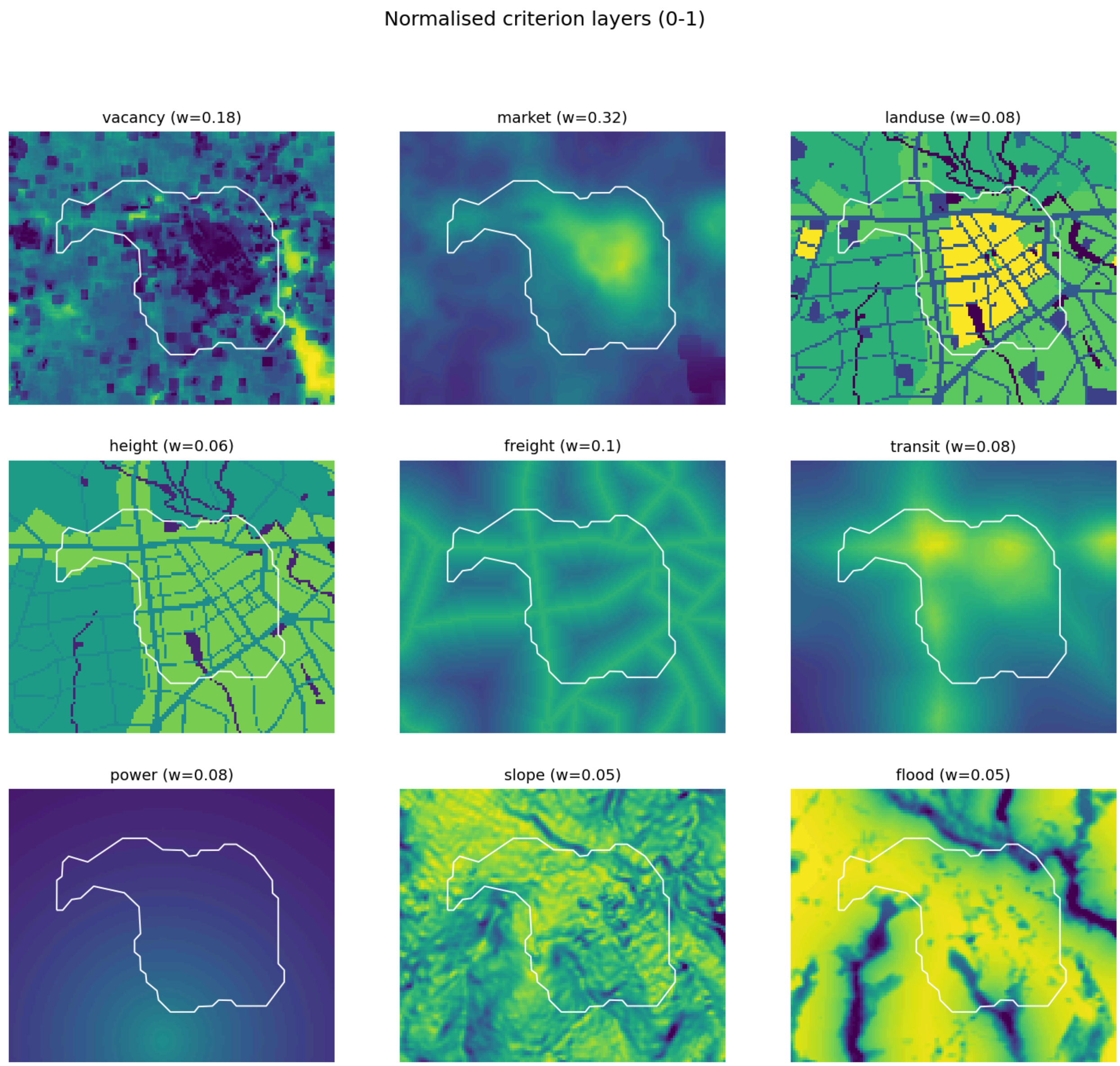
5. Method

Each criterion was normalised to 0–1 on a common 20 m grid (distance-decay for proximity criteria; category scores for zoning/height; min–max with percentile clipping for continuous layers), then combined by weighted linear overlay. Hard exclusions (score 0): heritage structures & historical sites, river buffers, parks/forest/cemetery, high-tension-line corridors, water reservoirs, and severe-drainage cells (top-15% risk). They cover 4.1% of the study area.

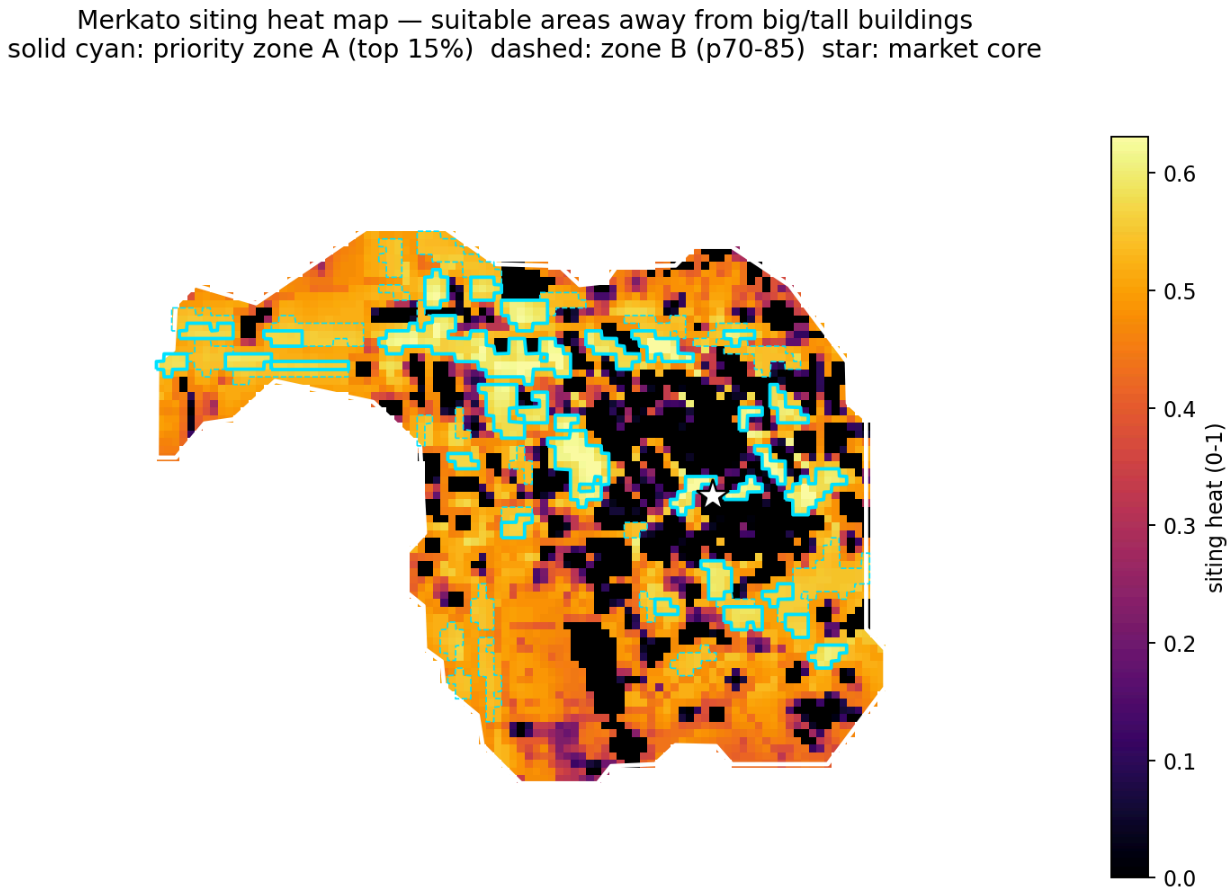
Criterion	Weight (base)	Weight (sensitivity)
Vacancy / low displacement	0.18	0.26
Trade gravity & catchment	0.32	0.24
Land-use compatibility	0.08	0.08
Height-regulation headroom	0.06	0.06

Road & freight access	0.10	0.10
Transit & footfall	0.08	0.08
Power capacity (proxy)	0.08	0.08
Slope & buildability	0.05	0.05
Flood / drainage risk (inverted)	0.05	0.05

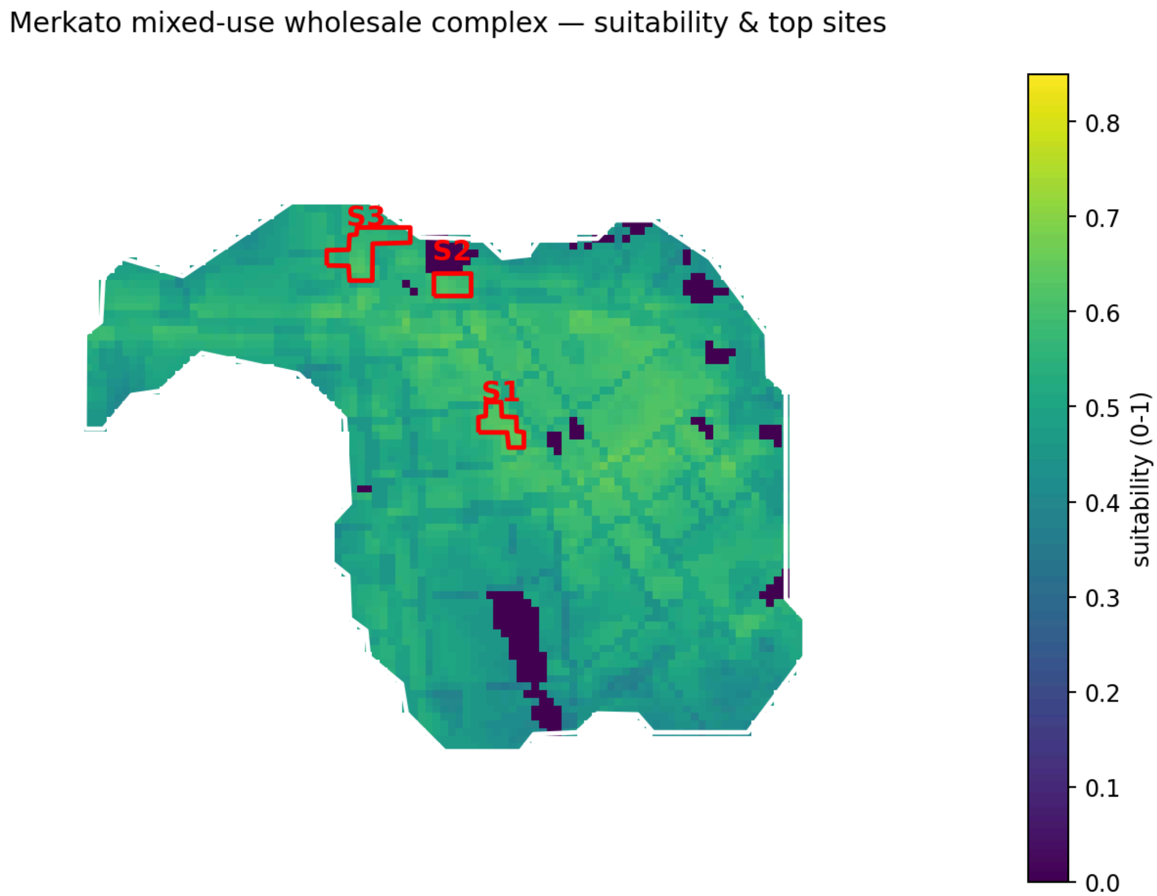
Weighting rationale. Trade gravity & catchment dominates (0.32) per explicit, twice-reaffirmed client direction — the complex must anchor in the market hotspot. The criterion itself is hotspot-centric: commercial-POI density leads (45%), core distance second (35%, sharp 350 m half-distance), population catchment third (20%). Vacancy/displacement remains second overall (0.18); freight access (0.10) and zoning (0.08) are the key operational and regulatory enablers; height headroom (0.06) captures the ~70 m massing; transit and power (0.08 each) support footfall and the data-centre/logistics load; slope and flood (0.05 each) are screening-level geotechnical proxies. Criteria 1 and 2 now pull in opposite directions by design (the hotspot is dense; vacancy prefers open fabric) — the height and big-building hard filters arbitrate. Final ranking applies a core-proximity multiplier (half-effect at 600 m), and candidacy requires the market criterion ≥ its in-boundary median (the hotspot rule); both are documented in scripts/config.py.



6. Results — siting heat map and priority zones



The heat map multiplies the weighted suitability surface by a satellite-measured height penalty (full value below 8 m, zero at ≥ 15 m), a big-building penalty (≥ 800 m^2 footprints, zero at 50% cell coverage) and the hard-exclusion mask. Dark voids are therefore precisely the areas the client asked to avoid — existing substantial/tall buildings — while bright ribbons are low-rise, hotspot-adjacent land. Zone A (top 15% of in-boundary heat, 30 polygons, 17.8 ha) and zone B (p70–85, 25 polygons, 13.7 ha) are delivered as `priority_zones.gpkg` with per-zone heat, market score, measured max height and core distance, for use in the groundwork and consent phase. The nearest zone-A pocket lies 58 m from the core anchor — small near-core pockets that a 5,000 m^2 parcel rule would discard remain visible in this framing, which is the point of deferring parcel selection.



6.1 Illustrative parcels (calibration only — not proposals)

To sanity-check that the zones can yield buildable parcels, the extraction rule was also run: suitability ≥ p45 within the boundary, **market criterion ≥ p50 (the hotspot rule)**, vacancy score ≥ 0.35, land-use score ≥ 0.45 (blocks road right-of-way, civic and green land), and a **substantial-structure filter**: buildings with footprints ≥ 800 m² are treated as significant regardless of missing height data — cells more than 20% covered by such structures are barred from candidacy, and no shortlisted site may have more than 10% of its area under them (all five have ≈ 0%). In addition, a **satellite-measured height filter** (Google Open Buildings 2.5D Temporal, June 2023): cells whose measured building height is ≥ 12 m (≈ G+3) are barred, and no shortlisted site may contain any structure ≥ 15 m — the shortlist's measured maxima are 7.5–9.5 m. Site polygons may consolidate minor internal lanes (single 20 m grid gaps, typical kebele lanes closed in Addis block redevelopment); arterial rights-of-way still separate sites, and hard blockers are never bridged. The scorecard reports each site's true built-coverage (from 131,701 footprints on a 5 m subgrid), big-building share, and mean/max measured height for audit. Every site was additionally verified visually on current high-resolution satellite imagery (§6.1). Note that genuinely empty land is nearly absent in the Merkato core: most shortlisted blocks carry dense *small-grain* (≤ ~780 m² footprint) single-storey fabric, which is precisely the redevelopment target — occupant surveys remain essential.

site_id	area_m2	suitability	suitability_alt	built_cover	big_cover	main_landuse	height_zone	vacancy	market	landuse	height	freight	transit	power	slope	flood
S1	7,455	0.62	0.59	0.60	0.00	Higher Level Market	Zone_2	0.38	0.74	0.96	0.78	0.43	0.69	0.21	0.73	0.85
S2	5,998	0.60	0.59	0.65	0.00	High Density Mixed Residence	Zone_2	0.44	0.60	0.75	0.80	0.60	0.83	0.16	0.76	0.82
S3	14,784	0.57	0.56	0.35	0.00	Parking Building	Zone_2	0.47	0.53	0.66	0.72	0.58	0.82	0.15	0.80	0.85

S1 — 7,455 m² (0.75 ha), centroid ≈ 9.03145°N 38.73576°E, 369 m from the Merkato core anchor. Dominant planned land use: *Higher Level Market*; height-regulation zone: *Zone_2*. Composite suitability 0.624 (alt-weights 0.595). Strengths: land-use compatibility, flood / drainage risk (inverted), height-regulation headroom. Weaker on: power capacity (proxy), vacancy / low displacement. Moderately built fabric — displacement must be verified block-by-block before this site is carried forward.

S2 — 5,998 m² (0.60 ha), centroid ≈ 9.03480°N 38.73459°E, 680 m from the Merkato core anchor. Dominant planned land use: *High Density Mixed Residence*; height-regulation zone: *Zone_2*. Composite suitability 0.599 (alt-weights 0.586). Strengths: transit & footfall, flood / drainage risk (inverted), height-regulation headroom. Weaker on: power capacity (proxy), vacancy / low displacement. Moderately built fabric — displacement must be verified block-by-block before this site is carried forward.

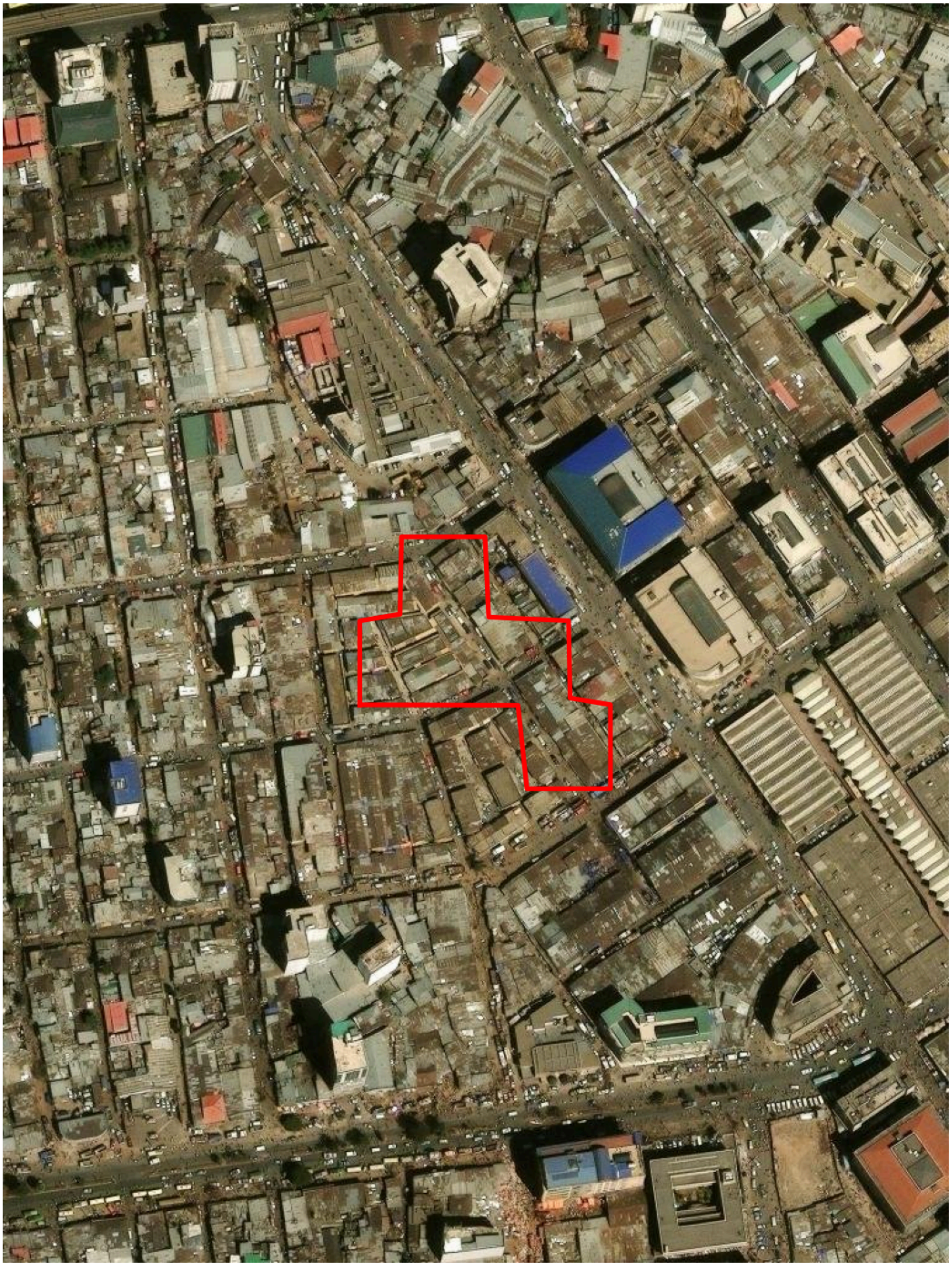
S3 — 14,784 m² (1.48 ha), centroid ≈ 9.03562°N 38.73252°E, 910 m from the Merkato core anchor. Dominant planned land use: *Parking Building*; height-regulation zone: *Zone_2*. Composite suitability 0.568 (alt-weights 0.563). Strengths: flood / drainage risk (inverted), transit & footfall, slope

& buildability. Weaker on: power capacity (proxy), vacancy / low displacement. Moderately built fabric — displacement must be verified block-by-block before this site is carried forward.

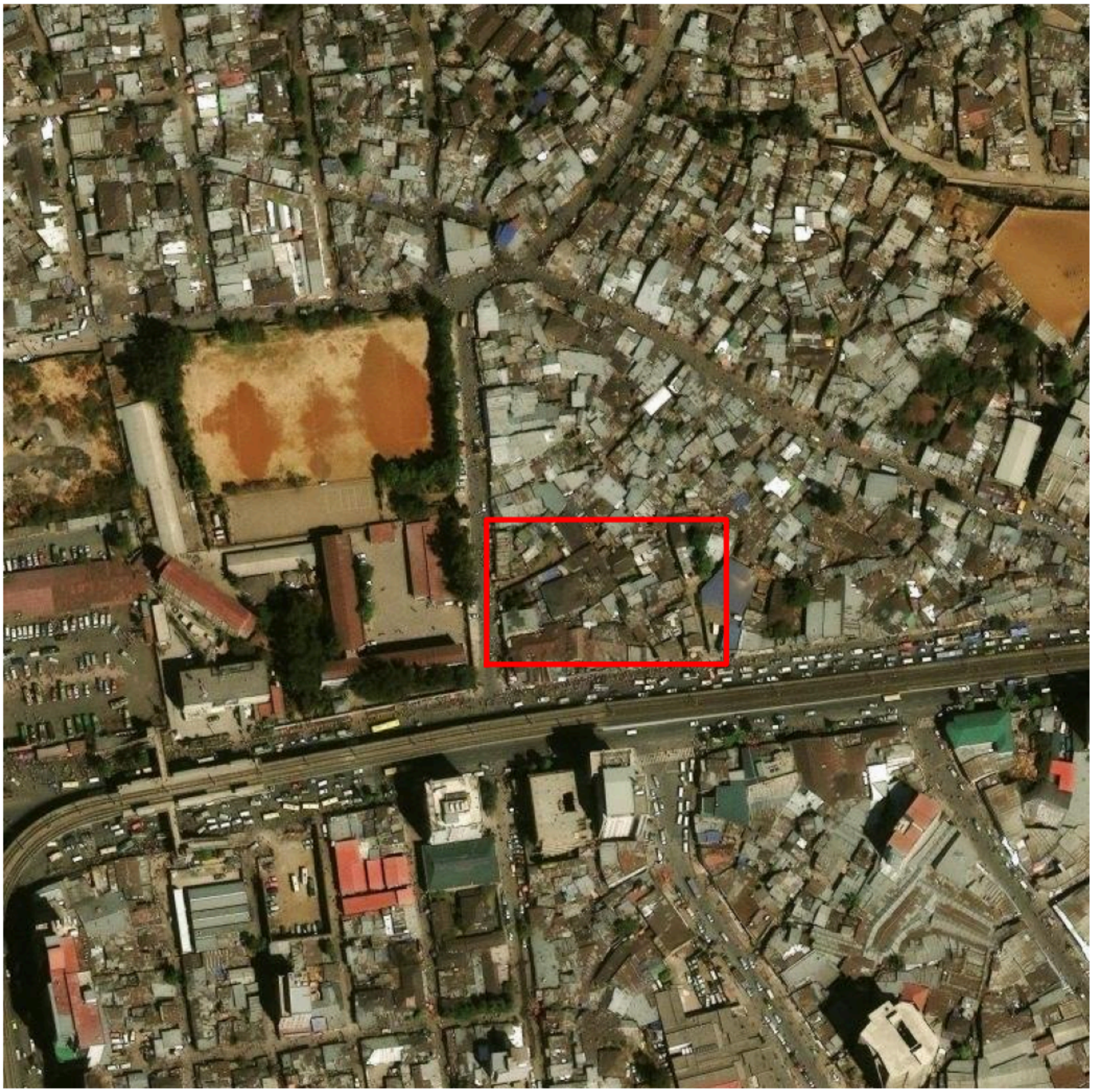
6.2 Satellite-imagery audit of the illustrative parcels

Each site was rendered on current Esri World Imagery (zoom 18) and inspected for tall or substantial structures. Findings: S1 — single-storey market-stall/shed fabric in the dense heart of Merkato, 369 m from the anchor; the surrounding mid-rise market halls all fall outside the outline. Dense trader occupancy — the strongest market position of any feasible parcel, and the heaviest relocation-planning burden. S2 — compact low-rise block fronting the arterial/LRT corridor beside a school compound (outside). S3 — a substantially cleared/bare parcel with rubble plus adjacent low sheds and a parking strip (35% built cover, emptiest of the three). No high-rise structure intersects any site outline.

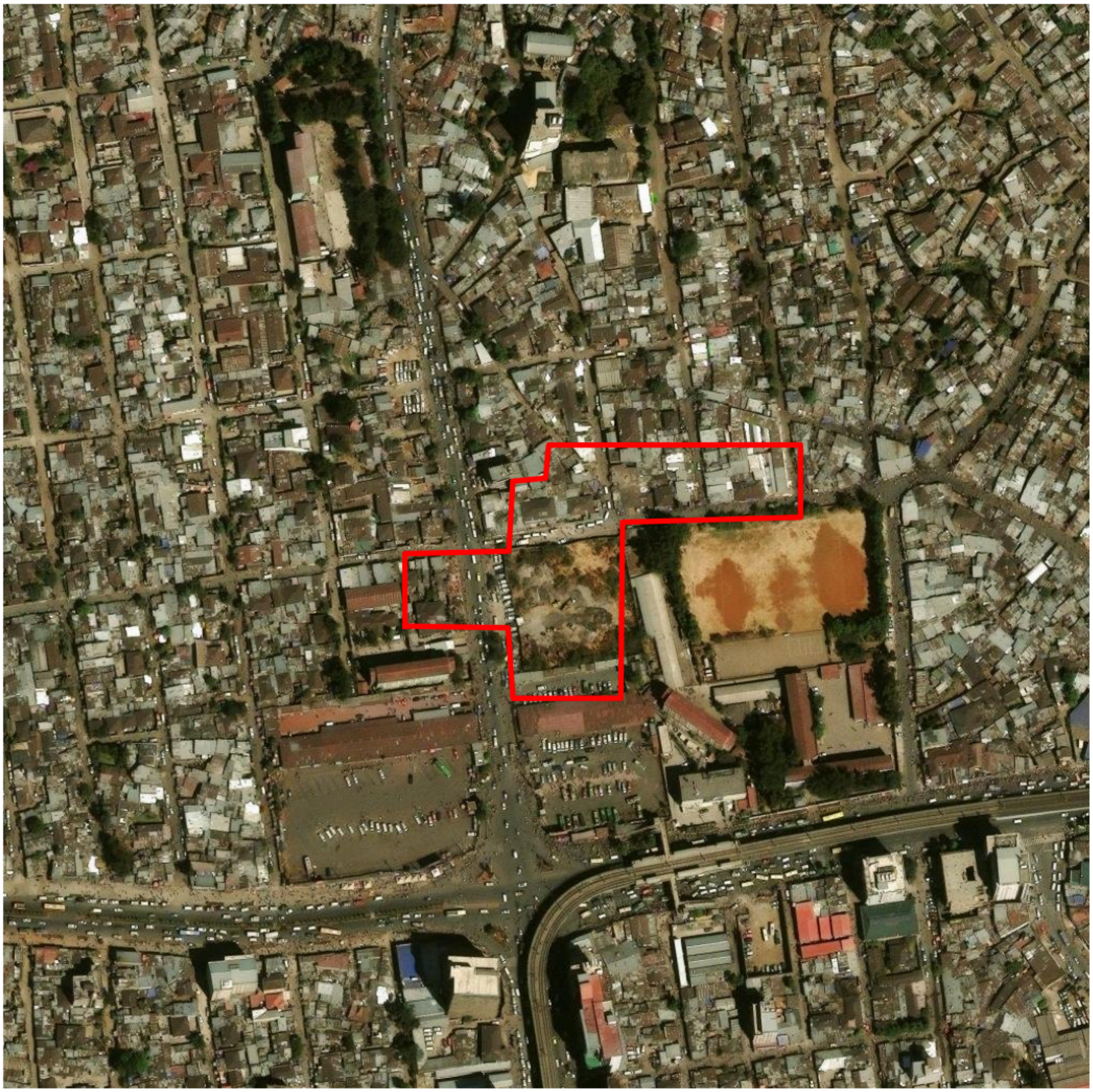
S1 — 7,455 m2 (Esri World Imagery, z18)



S2 — 5,998 m2 (Esri World Imagery, z18)



S3 — 14,784 m2 (Esri World Imagery, z18)



7. Sensitivity

A displacement-heavy alternative weighting (vacancy 0.26, market 0.24, landuse 0.08, height 0.06) was run in full. The two suitability surfaces correlate at r ≈ 0.983 and the same sites emerge — the shortlist is not an artefact of the chosen weights. Note the base weighting itself is the strongly market-forward set requested by the client; the alternative shows the displacement-first perspective.

8. Caveats and integrity statement

> This is a **planning-stage screening, not a legal determination**. Candidate sites are presented as *meeting the stated criteria, subject to the city's land-allocation authority* — not as a claim on specific land. Before any parcel is relied upon: verify the official cadastre and tenure status, obtain a zoning certificate, commission a geotechnical survey (3-basement excavation), confirm utility capacity with Ethiopian Electric Utility, and ground-truth the top candidates with site visits and an occupancy/displacement census.

- **Power criterion is a proxy**: OSM power data in the bbox is nearly empty (1 point); the criterion uses structural-plan power-station polygons and high-tension corridors. Absolute values are low everywhere; only relative differences are meaningful. EEU substation spare-load data must be obtained directly.
- **Flood criterion is a proxy**: a depression-depth surrogate (focal relative elevation) replaced full flow accumulation after a tooling failure; river-buffer proximity supplements it. A hydrological/drainage study is required for basements.

- **Building heights are satellite-measured, not surveyed**: per-cell maxima come from Google Open Buildings 2.5D Temporal (June 2023, model-derived from Sentinel-2 at ~4 m effective resolution, aggregated here to 20 m), supplemented by OSM tags (570 buildings) and a conservative ≥ 800 m^2-footprint assumption. The 2.5D model can under-estimate slender towers and reflects mid-2023 — anything built since is invisible to it, which is why the visual imagery audit (§6.1) and field verification of storey counts on shortlisted blocks remain required.
- **Height-regulation zones were scored by assumption** (Zone_1 highest cap → 1.0 … zone_4 → 0.35) because the legend PDF was not machine-readable; recovered MXD strings ("G+24=25", "G+6") support the ordering. Confirm zone caps with the Addis Ababa Plan Commission before design.
- Vacancy vs. market-gravity criteria overlap mildly; both favour the market core.
- WorldPop is a 2020 model surface resampled from 100 m; it informs relative catchment only.

9. Recommended next steps

1. Ground-truth the five sites (occupancy census, dilapidation survey, trader mapping).
2. Cadastre/tenure verification and land-allocation engagement with the city.
3. Geotechnical boreholes + groundwater assessment on the top two sites.
4. EEU load-flow enquiry for the data-centre node; telecom fibre availability.
5. Traffic/freight micro-simulation for the preferred site's service access.

Reproducible pipeline: `python run_all.py` — see README.md. All intermediate layers cached under data/; deliverables under outputs/.